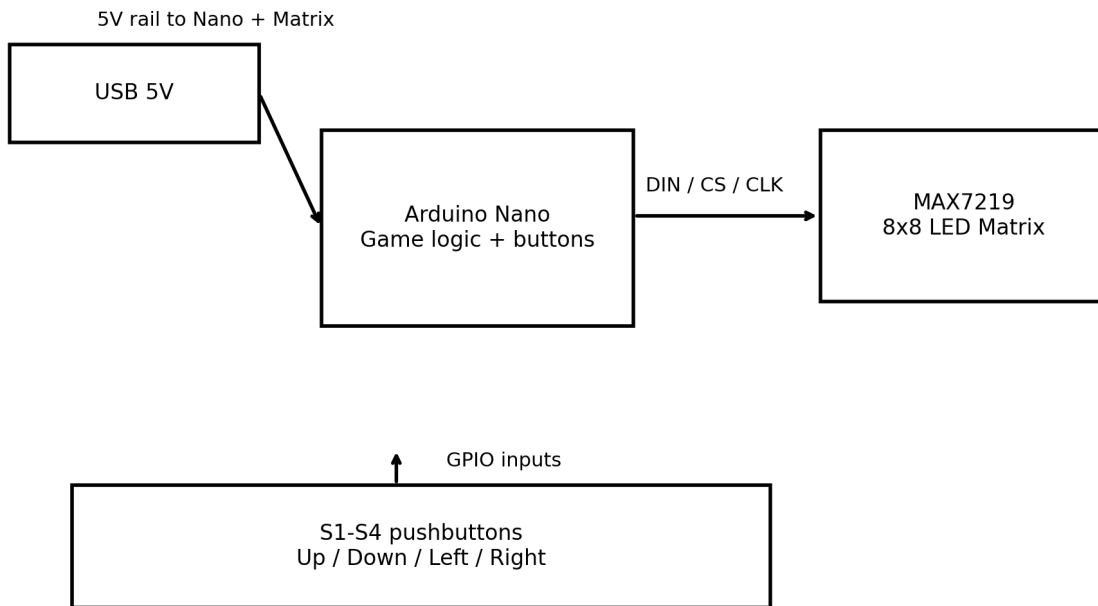


Snake Game LED Dot Matrix Circuit Package

USB-powered Arduino Nano compatible design using one MAX7219 8x8 LED matrix and four pushbuttons for directional control.

Assumptions: single 8x8 playfield, 5V USB power, active-low buttons using internal pull-ups, prototype-level design.

Files: schematic snake_matrix_schematic.svg, block diagram snake_matrix_block_diagram.png, firmware snake_game_arduino.ino



The detailed schematic is also included as an SVG file in the package directory. The block diagram above summarizes power, control, and display interconnects.

Item	Refs	Qty	Description	Unit \$	Ext \$	Supplier
1	U1	1	Arduino Nano compatible board (ATmega328P/328PB, USB-C preferred)	2.86	2.86	Gringo Gadgets
2	U2	1	8x8 LED dot matrix module with MAX7219 driver	1.25	1.25	Electropeak
3	S1-S4	4	Momentary tactile pushbutton 6x6 mm, 4-pin	0.35	1.40	DIYMORE
4	R1-R4	4	Resistor, through-hole	0.02	0.08	Estimated
5	C1-C2	2	Decoupling capacitor	0.03	0.06	Estimated
6	J1	1	USB-C breakout for 5 V power input (optional if Nano USB powers system directly)	2.95	2.95	Adafruit
7	H1-H3	1	Header pins / Dupont jumper set	2.00	2.00	Estimated
Total estimated cost					10.60	USD

Firmware summary: the Arduino sketch reads the four buttons, drives the MAX7219 with LedControl, updates snake position on a timed loop, and blinks the matrix on game-over.

Limitations: one 8x8 matrix makes the game very compact; prices were assembled from public web pages and some commodity parts use estimates rather than live cart-verified distributor quotes.